EXPERIMENTAL DATA LOG • THERMAL INSULATION

SPACESUIT THERMAL SHIELDING KIT

STUDENT LABORATORY TASK SHEET & EXPERIMENTAL REGISTER

STUDENT NAME(S):

TEAM / TEST IDENTIFIER:

DATE / LABORATORY PERIOD:

ASSIGNED TPS CONFIGURATION:

[ ] A: Bare [ ] B: Bubble wrap [ ] C: Mylar [ ] D: MLI Composite

PART A. EXPERIMENTAL TEMPERATURE LOG (0 TO 15 MIN)

60s INTERVALS

ELAPSED TIME	OBSERVED TEMP (T_{obs})	MODEL TEMP (T_{model})	RESIDUAL $\lvert\Delta T\rvert$ ($^\circ\mathrm{C}$)	OBSERVER FIELD NOTES & BOUNDARY STATUS
0 min	$80.0^\circ\mathrm{C}$	$80.0^\circ\mathrm{C}$	$0.0^\circ\mathrm{C}$	Capsule submerged; ice bath boundary verified.
1 min				Check immersion depth and probe position.
2 min				
3 min				
4 min				
5 min				Mid-phase checkpoint; verify ice slurry volume.
6 min				
7 min				
8 min				Record any change in bath condition.
9 min				
10 min				Check for leakage or probe contact with the wall.
11 min				
12 min				
13 min				
14 min				
15 min				Final reading; record bath temperature.

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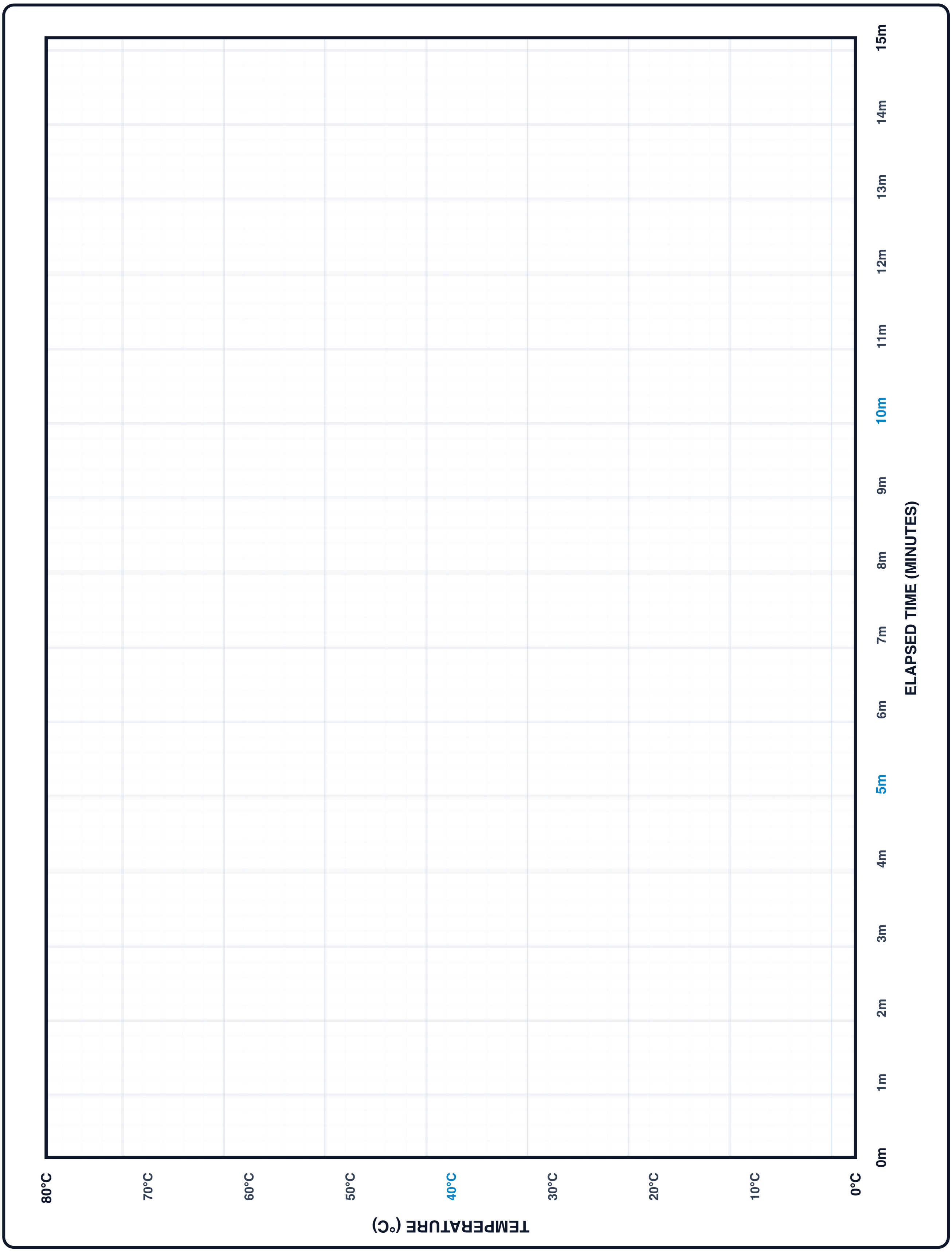
PART B. THERMAL COOLING CURVE GRAPH GRID (0 TO 15 MINUTES)

PART B. EXPERIMENTAL VS THEORETICAL COOLING CURVES

EMPIRICAL DATA (•) VS REFERENCE MODEL (---)

Plotting Instructions: Turn sheet 90° clockwise to plot. Plot observed temperature readings (T_{obs}) as solid points (●) and connect them with a smooth line. Plot the theoretical reference model (T_{model}) as a dashed curve (—).



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RESIDUAL, ERROR & UNCERTAINTY ANALYSIS

PART C. QUANTITATIVE RESIDUAL & ERROR ANALYSIS

STATISTICAL EVALUATION

1. Sum of Absolute Residuals: $\sum |\Delta T| = \sum |T_{\text{obs}} - T_{\text{model}}|$ $\sum |\Delta T| =$ ___________________ $^\circ\text{C}$

2. Mean Absolute Error (MAE): $\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |T_{\text{obs},i} - T_{\text{model},i}|$ $\text{MAE} =$ ___________________ $^\circ\text{C}$

3. Interpretation: Compare MAE with the thermometer accuracy and variation between repeated measurements.

Number of observations: --------------------

MODEL-AGREEMENT RESULT: **MAE =** _________ $^\circ\text{C}$ **[ ] uncertainty discussed** **[ ] model assumptions discussed**

CALCULATION WORKING AND ERROR DISCUSSION

Show residual calculations, MAE working, uncertainty sources, and any model assumptions used in your analysis.

PART D. POST-FLIGHT SCIENTIFIC QUESTIONS & ANALYSIS

Q1-Q5

Q1: Radiant Barriers vs. Direct Conduction

Reflective film can reduce radiative exchange when it faces a suitable gap. Why may it provide limited insulation when placed directly in ice water without a low-conductivity air or fibre spacer?

Q2: Understanding the Decay Constant (k)

In Newton's Law $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$, does a smaller k value mean better or worse insulation? Explain how k controls the cooling speed.

Q3: Why Multi-Layer Insulation (MLI) Works

Why can the multilayer configuration reduce the cooling rate relative to a single layer? Discuss trapped air, fluid motion, radiation and interfacial resistance.

Q4: Real-World Experimental Factors

What physical factors in the lab might cause your sensor readings to drift away from the computer simulation? Name at least two specific factors (e.g. sensor depth, stirring, seal leaks).

Q5: Ice Bath Temperature Drift

If the ice bath warms up from $0.0°\text{C}$ to $4.5°\text{C}$ during your 15-minute run, how should you update the environment temperature T_{env} in Mission Control so the model stays accurate?

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SCIENTIFIC REASONING AND WRITTEN RESPONSES

PART D. POST-FLIGHT SCIENTIFIC QUESTIONS & ANALYSIS

Q6-Q10

Q6: Effect of Capsule Water Volume

If you doubled the water volume from $100\ \mathrm{mL}$ to $200\ \mathrm{mL}$, how would that extra thermal mass ($Q = mc_p\Delta T$) change the cooling rate and the value of k?

Q7: Spacesuit Design Trade-Offs

Why can engineers not simply add an unlimited number of insulation layers to an EVA spacesuit? Discuss joint mobility, suit mass/bulk, and body heat buildup.

Q8: Specific Heat Capacity & Fluid Response

If water ($c_p \approx 4184\ \mathrm{J/(kg \cdot K)}$) were replaced with a liquid of lower specific heat capacity keeping mass identical, how would the cooling rate change?

Q9: Asymptotic Boundary Limits ($t \to \infty$)

According to Newton's Law $T(t) = T_{\mathrm{env}} + (T_0 - T_{\mathrm{env}})e^{-kt}$, what temperature value does the capsule approach asymptotically as time $t \to \infty$?

Q10: Telemetry Anomaly Diagnostics & MAE

If measured capsule temperature drops significantly faster than predicted, what physical anomaly is the most plausible explanation, and how can MAE be reduced?

STUDENT SIGNATURE: ________________ INSTRUCTOR VALIDATION: ________________ EVALUATION SCORE: ________________ / 100